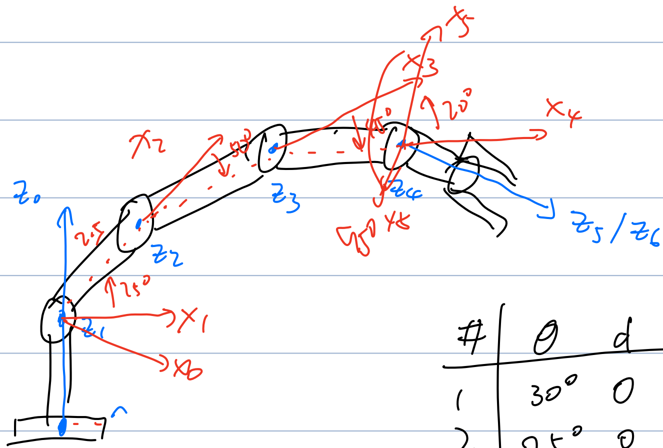
T3.

Q1.



#	θ	d	a	α
1	30°	0	0	90°
2	25°	0	2.5	0°
3	-50°	0	2.1	0°
4	-45°	0	1.8	0°
5	20°	0	0	90°
6	25°	0	0	0°

Q2. ${}^{T6}J \, T = F \Rightarrow T = {}^{T6}J^{-1} F = \underbrace{{}^{T6}J^{T} F}$

, where ${}^{T6}J = \begin{bmatrix} \quad \end{bmatrix}$

joint 1: revolute

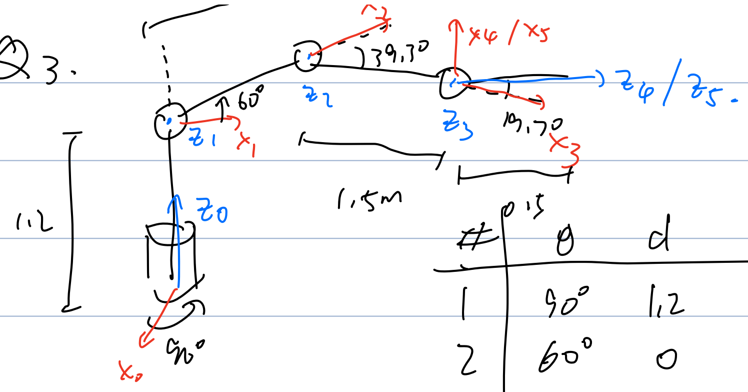
$${}^{T6}J_1 = \begin{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} \\ n_z \\ o_z \\ a_z \end{bmatrix}$$

$A_1 A_2 A_3 A_4 A_5 A_6$.

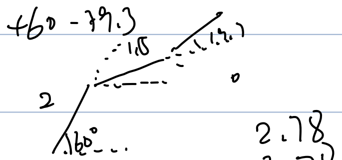
. . . . . . too complicated !!!

2

Q3.



#	θ	d	a	α
1	$90°$	1.2	0	$90°$
2	$60°$	0	2	$0°$
3	$-39.3°$	0	1.5	$0°$
4	$19.7° + 90°$	0	0	$90°$
5	0	0.5	0	$0°$



2.78
3.78

Q5. a.

$$K_1 = \frac{1}{2} \cdot \frac{1}{3} m_1 l_1^2 \dot{\theta}^2 \quad \Rightarrow \quad K = K_1 + K_2$$

$$K_2 = \frac{1}{2}\left(\frac{1}{12} m_2 l_2^2 + m_2 r^2\right) \dot{\theta}^2 + \frac{1}{2} m_2 \dot{r}^2$$

b. $$P = P_1 + P_2 = m_1 g \frac{l_1}{2} S\theta + m_2 g r S\theta$$

c. $$L = K - P$$

d. $$T = \frac{\partial}{\partial t}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta}$$

$$F = \frac{\partial}{\partial t}\left(\frac{\partial L}{\partial \dot{r}}\right) - \frac{\partial L}{\partial r}$$

$$\frac{\partial L}{\partial \dot{\theta}} = \frac{1}{3} m_1 l_1^2 \dot{\theta} + \left(\frac{1}{12} m_2 l_2^2 + m_2 r^2\right) \dot{\theta}$$

$$\frac{\partial}{\partial t}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = \left(\frac{1}{3} m_1 l_1^2 + \frac{1}{12} m_2 l_2^2 + m_2 r^2\right) \ddot{\theta} + 2 m_2 r \dot{r} \dot{\theta}$$

$$\frac{\partial L}{\partial \theta} = -\left(m_1 g \frac{l_1}{2} + m_2 g r\right) \cos\theta$$

$$\frac{\partial L}{\partial \dot{r}} = m_2 \dot{r}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right) = m_2 \ddot{r}$$

$$\frac{\partial L}{\partial r} = m_2 r \dot{\theta}^2 - m_2 g S_\theta$$

$T = \ldots$

$F = \ldots$

$$\begin{bmatrix} \tau \\ F \end{bmatrix} = \begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix} \begin{bmatrix} \ddot{\theta} \\ \ddot{r} \end{bmatrix} + \begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix} \begin{bmatrix} \dot{\theta}^2 \\ \dot{r}^2 \end{bmatrix} + \begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix} \begin{bmatrix} \dot{\theta}\dot{r} \\ \dot{r}\dot{\theta} \end{bmatrix} + \begin{bmatrix} \quad \\ \quad \end{bmatrix}$$